
Report: **AM2024-0074**

TEM investigation on particle samples of natural origin

Short Summary 4 – Rev. 1

Client:

Curia Germany GmbH
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65926 Frankfurt am Main

Contractor:

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Datum: 2024.10.15 08:14:10 +02'00'

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Documentation Overview

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ISC Offer Number

2210-2024-51000-45-02801-00010 and
2210-2024-51000-45-02801-00019

Order of the Client

signature on offer 2210-2024-51000-45-02801-00010
dated 26 July 2024 and
signature on offer 2210-2024-51000-45-02801-00019
dated 09 September 2024

Place of Performance and Processing Time

Place of Performance

Center for Applied Analytics
Fraunhofer Institute for Silicate Research
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Processing Time

Sample receipt	28.08.2024
Begin of the analysis	19.09.2024
End of the analysis	10.10.2024
Date of report	14.10.2024

Sample Overview

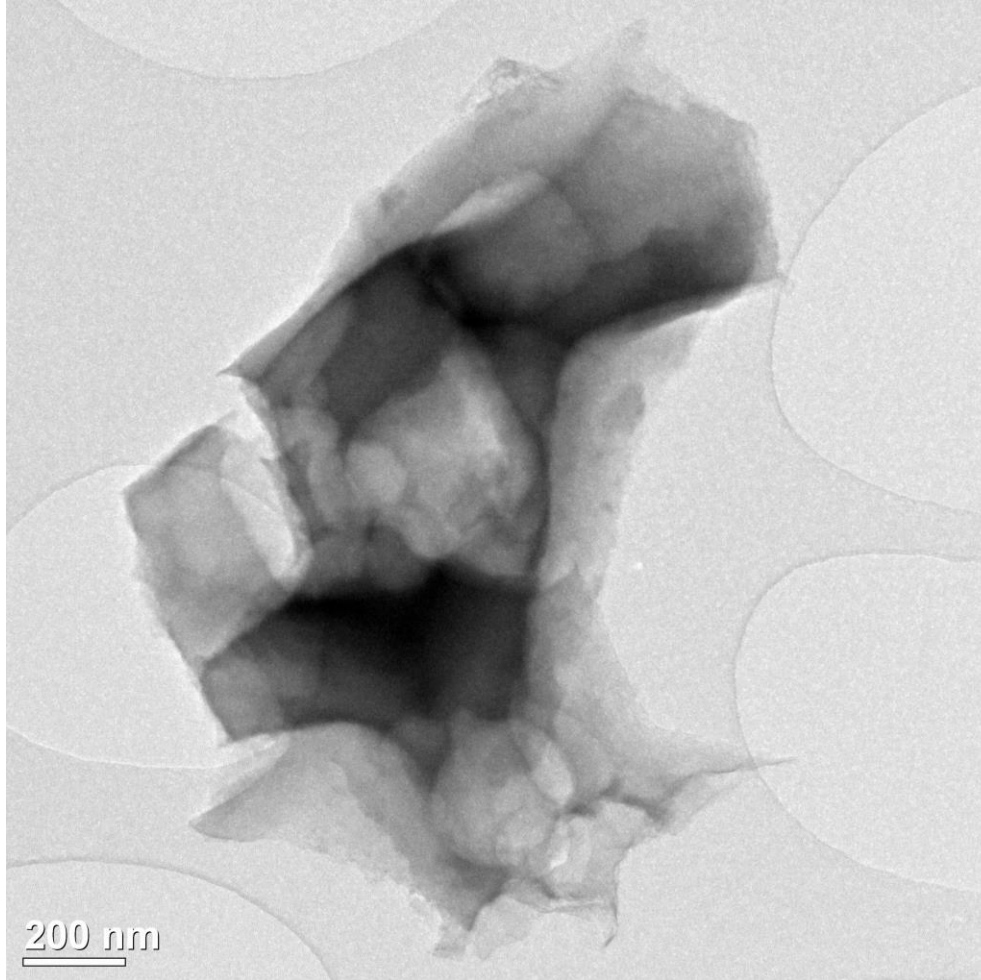
Sample No.	Description
P03	Graphene – Batch: MB2976A – Date: 23.08.2024

Procedure of Testing

- The received sample was ground before a small amount of the powder was dispersed in 1 part IPA and 4 parts of water.
 - The dispersion was treated with an ultrasonic finger for 3 min.
 - The dispersion was dripped onto a Lacey Carbon TEM grid covered with ultra thin foil (EMR ultra-thin carbon) suitable for high resolution imaging.
 - The sample was dried on air without removing any excess liquid and afterwards transferred into the SEM. There was no sputter coating applied.
 - The investigation of the particles was done after prior SEM imaging (see short summary 3) in TEM brightfield mode.
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- **Note / Reason for revision:** During the investigation it turned out that the carbon grid itself showed weak diffraction patterns. Thus, the results of the diffraction analysis were retracted. Those investigations are redone as fast as possible.

TEM investigation

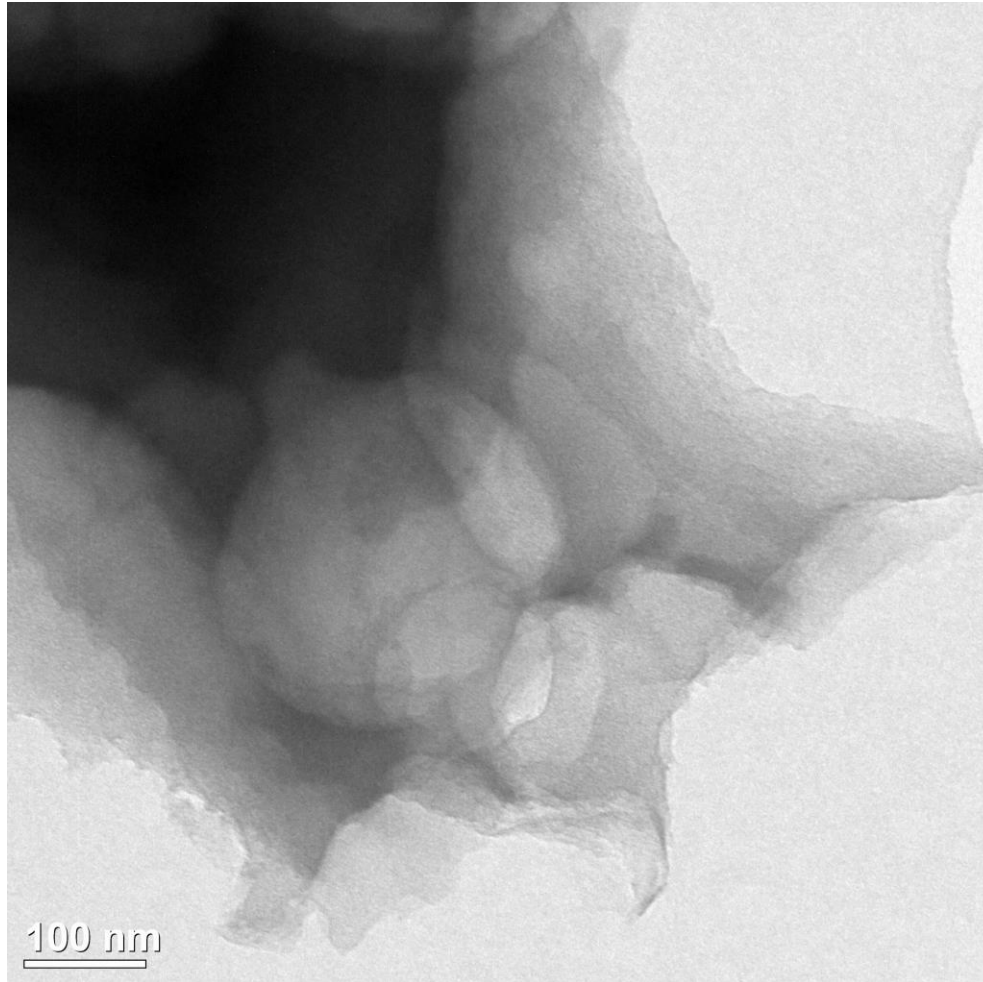
Sample P03 (MB2976A)



- This TEM brightfield image shows a larger particle of sample P03 at a magnification of 10.000 times.
- A porous structure especially in the lower part can be seen.

TEM investigation

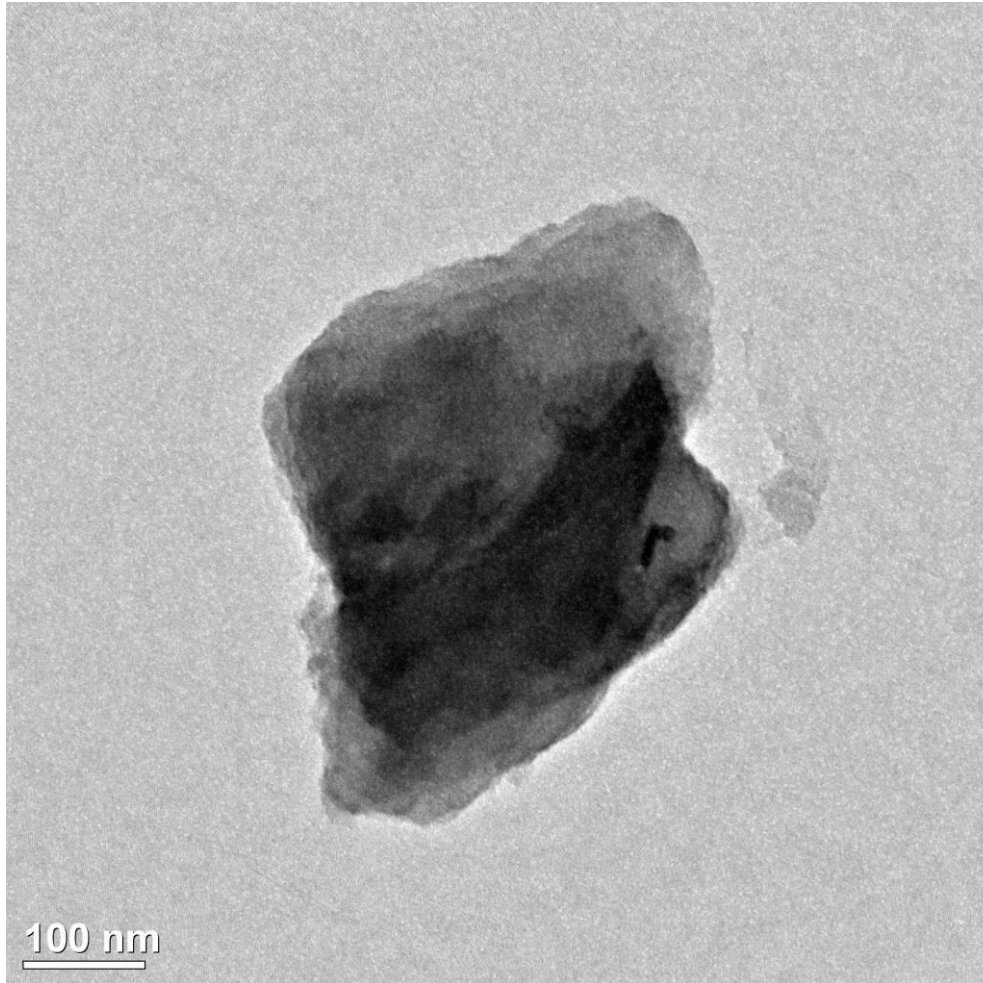
Sample P03 (MB2976A)



- This image shows the lower, porous part of the particle at a magnification of 25.000 times.

TEM investigation

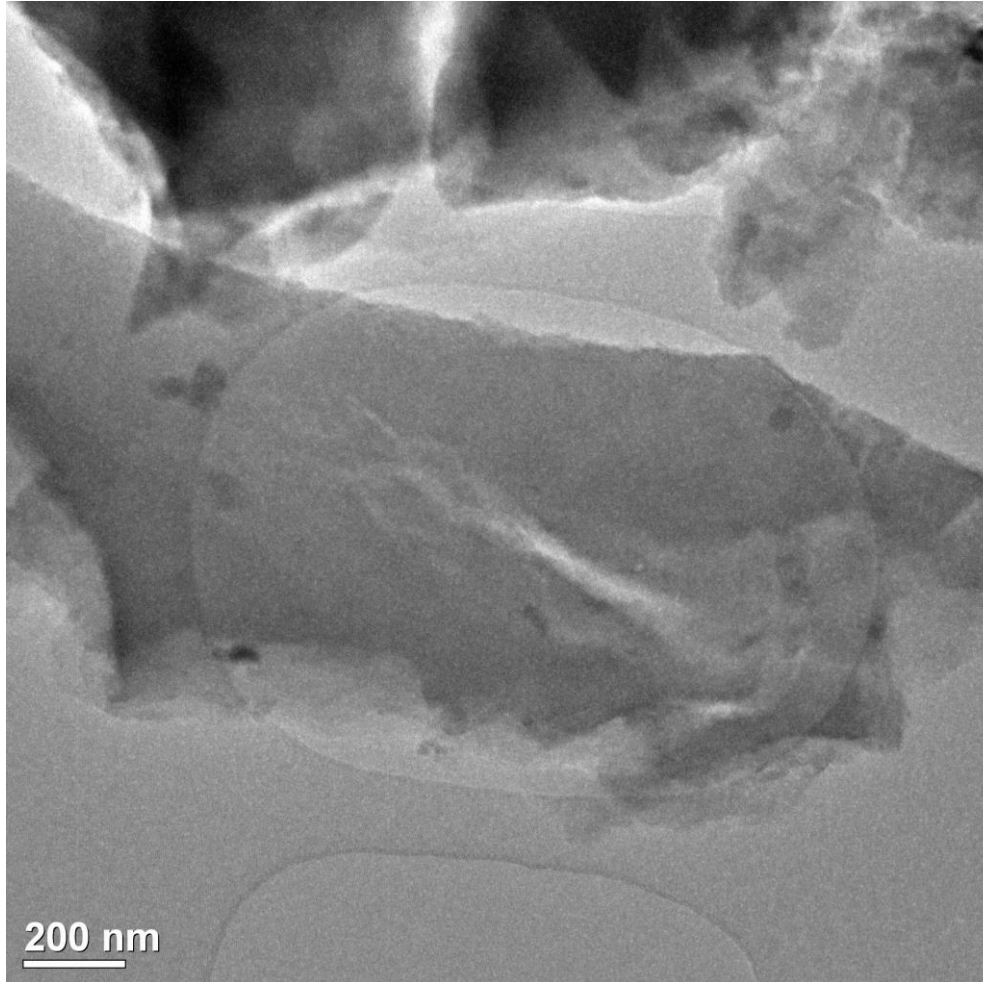
Sample P03 (MB2976A)



- This image shows a smaller particle at a magnification of 25.000 times.
- This particle appears to be compact.

TEM investigation

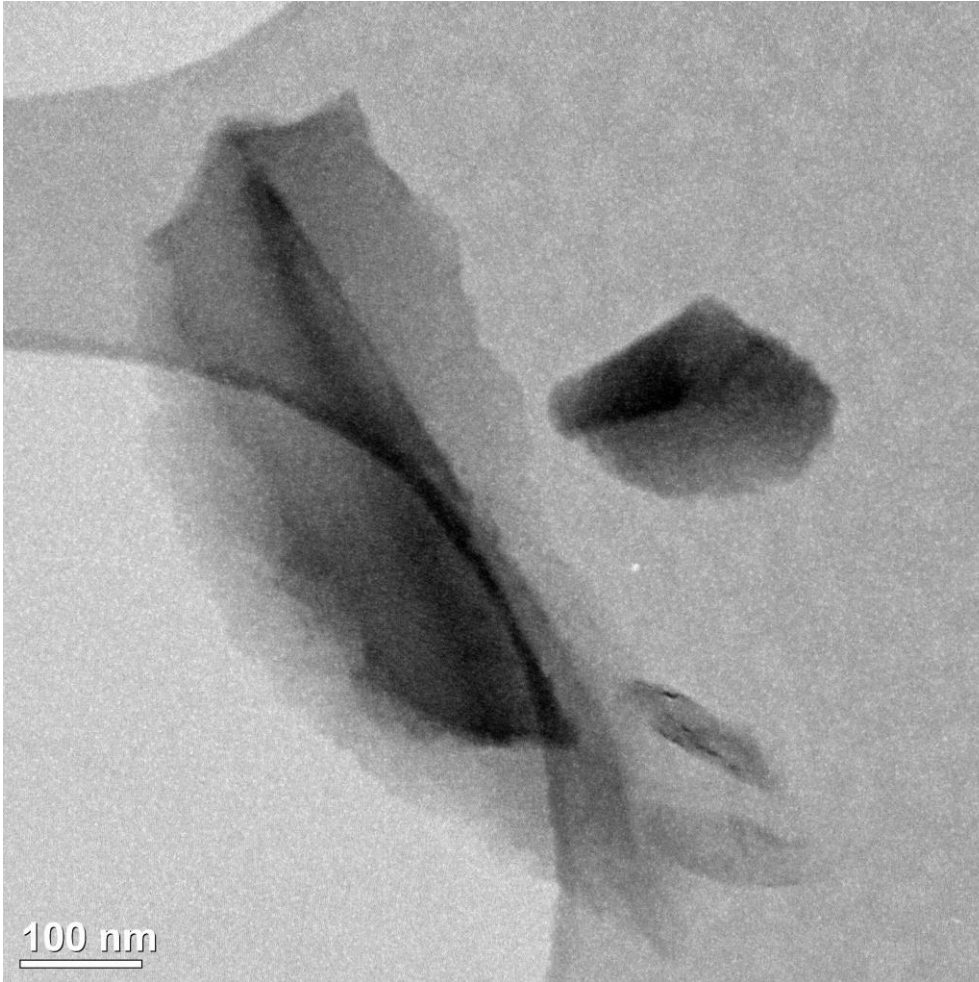
Sample P03 (MB2976A)



- This image shows a plate-like particle at a magnification of 10,000 times.
- This particle appears to be rather compact.

TEM investigation

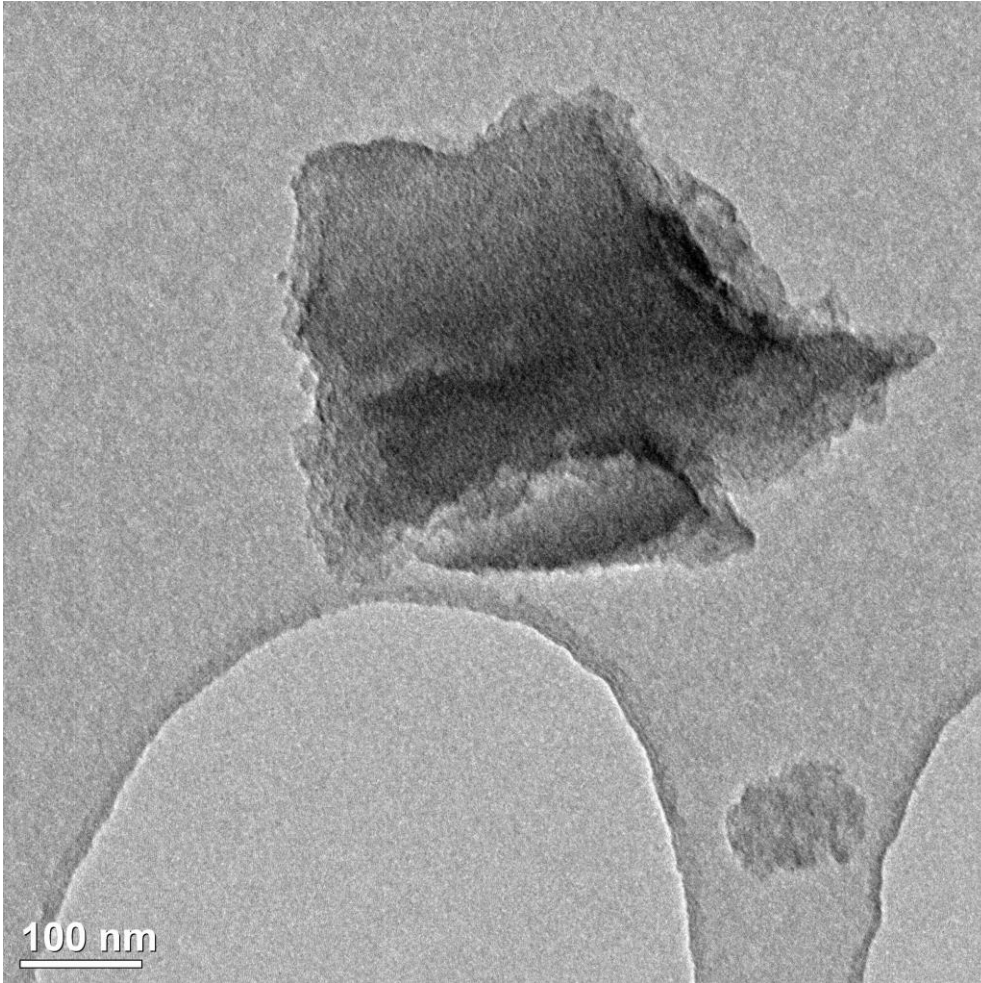
Sample P03 (MB2976A)



- This TEM image shows a plate-like particle on the left side and a compact one on the right side.

TEM investigation

Sample P03 (MB2976A)



- This TEM image shows two particles, a bigger one in the upper part and a smaller one in the lower part.

Final Remarks

- The analyses are carried out by specifically qualified employees and by the use of up-to-date analysis machines. Nevertheless, we are not able to provide guarantee for the correctness of the drawn conclusion.
- The results only refer to the provided and examined samples.
- Sample taking and shipping of the samples to the ZAA took place under the responsibility of the client.
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End of report AM2024-0074 – Short Summary 4 Rev. 1

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